primepayments

MERCHANT API DOCUMENT

DEPOSIT/FUND IN

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Document Release Note

Version	Date	Description	Author
1.0.0	31 Aug 2020	First version	K5
1.0.1	30 Oct 2020	Update the changes according to the latest upgrade in API	K5
1.0.2	27 Jan 2021	Add Java version of Message Encryption Example	K5
1.0.3	29 Jan 2021	Updated Testing API endpoint	K5

Prerequisite

Merchant Account

When merchant account is ready, we will provide merchant the followings

1) Merchant Back Office URL
2) API Endpoint URL
3) Merchant ID (16 characters, alphanumeric unique ID to identify merchant)
4) Merchant Key (16 characters, secret key for encryption)
5) Merchant Hash IV (16 characters, the hash IV (Initiation Vector) used for encryption)

Fund In (Deposit) Request

Test Environment
Merchant Back Office : http://merchant-test.primepayco.com/
API Endpoint : http://api-test.primepayco.com/api/merchant/transact
HTTP Method : POST
HTTP Header : Content-Type: application/json
Request Data Format : JSON
Response Data Format : JSON

Request Parameters

Field ID	Field Name	Data Type	Description	Example Value
merchantId		Char (16)	This will be provided	
message			Refer Message details in the table below. The message in json format is encrypted json data, refer the **Appendix A - Message** Encryption section below for the data fields	

Message (in JSON Format)

Field ID	Field Name	Data Type	Description	Example Value
Mandatory				
merchantId		Char (16)	This will be provided	
transactionId		Char (45)	A unique reference from merchant to identify the fund In/deposit request transaction.	
transactionType		Char (1)		Possible Value: **D**: Deposit

Field ID	Field Name	Data Type	Description	Example Value
payMethod		Char (45)	Determine the flow of deposit **offline** : customer deposits by copying bank details, required about 5-10 minutes approval time. **ebank** : customer requires to login using his/her username and password of the selected bank to deposit directly, instantly approval.	Options: **offline** **ebank**
currency		Char (3)	Currency Code (ISO 4217)	MYR
amount		Decimal (20, 2)		
callback		URL	When request is completed, a confirmation request will be sending to the callback URL	
response		URL	Need to add a query parameter success=:status Possible value for :status 1: Payment Success 0: Payment Failed	http://domain.com /response?success=:status
Optional Fields				

Field ID	Field Name	Data Type	Description	Example Value
lang		Char (2)	Default language for the content of redirect, default is English	Values: en: English cn: 中文(简体) th: Thai
designatedBankCode		Char (20)	Selected Bank, leave this empty the redirect URL will display bank selection instead of bank details for deposit.	
bankName		Char (45)	Sender Bank Details	
bankAcctName		Char (45)		
bankAccNum		Char (45)		
bankBranch		Char (45)		

Example

Body Data (JSON Format)


```
{
	"merchantId":"94b2522ba609909d",
	"message":"sflGy561H0rP/C+JvyUuCZ2W6k3VlxZrUWiy8TMqqpvOvoqYI2CSoJOIx2ogyyfoeonlounPuFDWQfDuQyH7RirMMLaFpuO0e90C9l
QDTeon/Cnk4L0ZfNdbYgjRKrS42dNWGyKV5kSy8FqCI5DJSE/cKJUzmNuy6A/8n+H7MaIQPprUV1jwv7eLZo/ls++rBLz0OCkeJFs/Tc+iJN1ZubOBRiCFK
jpZbfNkrhEqYfa4BIFvsGhvldwhza6SXAVyF4VSnUSmX7hcCA2Hmz6V7yqy5y2sUqmNDj2hfpjM220YTGaFIiEIjyfNW8tE4ZsUDube273JhLmpGByYCy9w
Ww=="
}
```


Message before encrypted


```
{
	"merchantId": "94b2522ba609909d",
	"transactionId": "20201030023201612",
	"transactionType": "D",
	"payMethod": "offline",
	"currency": "RMB",
	"amount": "500.00",
```

```
      "callback": "https://enmu1r0iut2ws.x.pipedream.net",
      "response": "https://enmu1r0iut2ws.x.pipedream.net?success=:status",
}
```

Response Parameters

Field ID	Field Name	Data Type	Description	Example Value
status		Char (10)		success or fail
When status = success				
referenceId		Char (45)	The unique fund out/withdrawal request transaction ID from us for your reference	`4acfa29f-41e4-4f3f-af6c-5e8ff844e722`
referenceNo		Number	The unique friendly request transaction no from us for your reference	248
redirectUrl		URL	URL to display the bank account details to deposit.	
designatedBankRemark		Char (12)	For customer to enter as Bank Reference expected to be enter in deposit	
designatedBank			Selected bank to deposit	
bankName		Char (45)	Selected bank name	
bankBranch		Char (45)	Selected bank branch	
bankAcctName		Char (45)	Selected bank account holder name	
bankAcctNum		Char (45)	Selected bank account number	
dateExpired		Date Time	Expiry date of the deposit request	
deposit_amount		Decimal (20, 2)	Expected amount to deposit	
When status = fail				
errors		Text	When 1) Incorrect Parameters	When status = fail

Field ID	Field Name	Data Type	Description	Example Value
			2) There is no bank account available.	

Example

Response Data when the request is success


```
{
	"status": "success",
	"referenceId": "7712ce58-1201-4be0-be7e-f0e2a89f9364",
	"referenceNo": 314,
	"redirectUrl": "http://uat.bank.sdgaming.net/pay/c9b035bdebb5606b9b342304314820d5",
	"designatedBankRemark": "3141111",
	"designatedBank": {
		"bankName": "华夏银行",
		"bankBranch": "",
		"bankAcctName": "测试名二",
		"bankAcctNum": "2222222222"
	},
	"dateExpired": "2020-10-30 14:37:02",
	"deposit_amount": "500"
}
```


Response Data when request is failed


```
{
	"status": "fail",
	"errors": "There is no bank account available."
}
```

Callback URL Request

We will inform merchant about the status change of the fund in/deposit request via the Callback URL. We may send the request multiple times and Merchant is required to handle the following situations, when fund in/deposit request is

1) **transferring** – the requested fund in/deposit amount is in the progress of verifying
2) **completed/success** – the requested fund in/deposit amount is successfully transferred to the designated bank account, but there could be the following situations:
    a. completed with adjustment – service fee could be adjusted by us due to error input.
    b. completed and refund later – due to wrongly identified the deposit transaction
3) **rejected** – due to incorrect designated bank details or requested by merchant
4) **failed** – after N attempts of verification but cannot find the deposit transaction in the designated bank account
5) **expired** – after a given expiry time to deposit

HTTP Method : POST
HTTP Header : Content-Type: application/json
Request Data Format : JSON
Response Data Format : JSON

Request Parameters

Field ID	Field Name	Data Type	Description	Example Value
merchantId		Char (16)	The unique merchant account id	
transactionId		Char (45)	The unique reference from merchant to identify the fund in/deposit request transaction.	
message		Text	Refer Message details in the table below. The message in json format is encrypted json data, refer the	

Field ID	Field Name	Data Type	Description	Example Value
			Appendix B - Message Decryption section below to decrypt the string in order to obtain the json format data.	

Message (in JSON Format)

Field ID	Field Name	Data Type	Description	Example Value
Mandatory				
referenceId		Char (45)	The unique fund out/withdrawal request transaction ID from us for merchant reference	
status		Integer	The request status	Possible values **1**: Completed **2**: Rejected **4**: Refunded **5**: Failed **6**: Expired
statusDesc		Char (20)	The description of the request status	When status value is 1 - **Completed** 2 - **Rejected** 4 - **Refunded** 5 - **Failed** 6 - **Expired**
requestDate		DATETIME	The fund in / deposit request date Date format: YYYY-MM-DD HH:MM:SS	2020-05-16 18:01:08
transactionType		Char (1)	The transaction type	Possible Value: **D**: Deposit
transactionTypeDesc		Char (20)	The description of the transaction type	When transaction type is D - **Deposit**

Field ID	Field Name	Data Type	Description	Example Value
currency		Char (3)	Currency Code (ISO 4217). As per the currency from the request	MYR
amount		Decimal (20, 2)	As per the amount from the request	1000.00
transactionDate		DATETIME	The date of status changed Date format: YYYY-MM-DD HH:MM:SS	2020-05-16 18:01:08
remark		Char (255)	The remark input by us	
When status = "28" (Transferring)				
fromBankAcctName		Char (45)	Depositor Name	
transactionAmount		Decimal (20, 2)	The actual amount of the fund transferred	
serviceFee		Decimal (20, 2)		
When status = "1" (Completed)				
designatedBank			Selected bank to deposit	
bankName		Char (45)	Selected bank name	
bankBranch		Char (45)	Selected bank branch	
bankAcctName		Char (45)	Selected bank account holder name	
bankAcctNum		Char (45)	Selected bank account number	
transactionAmount		Decimal (20, 2)	The actual amount of the fund transferred	
serviceFee		Decimal (20, 2)		
adjustment			When it is completed but there is input error of service fee/transaction fee. Refer to the Adjustment section below	
When status = "4" (Refunded)				

Field ID	Field Name	Data Type	Description	Example Value
refundedReason		Char (255)	The reason of the reverse of the fund out.	

Adjustment (in JSON Format)

Field ID	Field Name	Data Type	Description	Example Value
Mandatory				
adjustedDate		DATETIME	The date that we make the adjustment Date format: YYYY-MM-DD HH:MM:SS	
adjustedReason		Char (255)	The reason of the adjustment	
Conditional				
serviceFeeBeforeAdjusted		Decimal (20, 2)	This will return the previous value of the service fee when the adjustment is on service fee. The new value is replaced in the **serviceFee** above.	

Example

Request Post Data


```
{
  "merchantId": "82164414b30aa575",
  "transactionId": "20200903075832695",
  "message":
"Eis/kISbtVtkLckK/LHMQTCEFGr787ymoejvM0m8FFhe0vOJvFNWyH1b/nn3QOc3RlTEbtmuf5qn35bXVuUeQUYyuPfeZzvcfOnjILee/k+xyhgErwC+5F
eyi6c1/sqDXymJN5oUV8avWI/S752GpvzBeqsx3Y8Om+DKBkMJqgT2T/gudc4/T/5jE9ZvRhNiN3rdsLFwmLNUhLhXA1kqiUGnwFF8pEnH1ApUMEirD3FfY
vY2a7vYB6Xz2sI755KtDvLXO81SWDj3VNFQTgB8F01JMvLWoxjx+9/VAp1sfMbr4mcFIS2rmWgIOSbjTchyMb5V0JXlHLOkxHHVfBlH3jkiyzwenu6rC6mc
1wcxBxc4WdKAFs1N12ZanarTrq6AXPg/MHiHKuajwVyEkCjDJQLeUf1oki3jpeFZLal2HaiWEEhFuqAWYq0I7y9vnQIzjuVG6PFgEwXX63XDA5qmX6KKQng
6VGMjj7n/ftMdiIiAZnMSZK4Fv5GtWzw6YMUj7+g4OwfGtqqKmIaamD9TpKcQDJ7qZMox4awWEviwtvIdDaRLQl2HdxvX5u28hfgI"
}
```


Decrypted Message:


```
{
  "referenceId": "dbeb1853-d10c-44d4-ab79-af6fb5025ae5",
```

```
  "status": 1,
  "statusDesc": "Completed",
  "requestDate": "2020-08-27 19:18:45",
  "transactionType": "D",
  "transactionTypeDesc": "Deposit",
  "currency": "THB",
  "amount": "500.00",
  "transactionDate": "1970-01-01 07:30:00",
  "remark": "",
  "designatedBank": {
    "bankName": "Krung Thai Bank",
    "bankBranch": "",
    "bankAcctName": "TEST",
    "bankAcctNum": "4444444444"
  },
  "transactionAmount": "500.12",
  "serviceFee": "7.50"
}
```


Response Parameters

Field ID	Field Name	Data Type	Description	Example Value
status		Char (10)		**success** or **fail**
message		Char (255)	Description for Merchant reference	

Example

Response JSON


```
{
  "status":"success",
  "message":""
}
```

Bank List

Test Environment
API Endpoint : http://api-test.primepayco.com/api/merchant/banklist
HTTP Method : POST
HTTP Header : Content-Type: application/json
Request Data Format : JSON
Response Data Format : JSON

Request Parameters

Field ID	Field Name	Data Type	Description	Example Value
merchantId		Char (16)	This will be provided	
message			Refer Message details in the table below. The message in json format is encrypted json data, refer the **Appendix A - Message** Encryption section below for the data fields	

Message (in JSON Format)

Field ID	Field Name	Data Type	Description	Example Value
Mandatory				
merchantId		Char (16)	This will be provided	
payMethod		Char (45)	Determine the flow of deposit **offline** : customer deposits by copying bank details, required about 5-10 minutes approval time. **ebank** : customer requires to login using his/her username and password of the	Options: **offline** **ebank**

Field ID	Field Name	Data Type	Description	Example Value
			selected bank to deposit directly, instantly approval.	
currency		Char (3)	Currency Code (ISO 4217)	MYR
amount		Decimal (20, 2)		

Example

Body Data (JSON Format)


```
{
  "merchantId":"82164414b30aa575",
  "message":
"N+uVCc2MrpWk+rr2kaQ1pXyZqr4uS4v0Rhk7TsfezgTwsWUXmq4bbgcI10ojXSfoCun9Tll2CfceBuEw/joYHLoCgssNFaPgfM6NdJ8/N5Jw1HQpfGToYeeeWJ3n3fds"
}
```


Message before encrypted


```
{
  "merchantId": "82164414b30aa575",
  "payMethod": "offline",
  "currency": "THB",
  "amount": "200"
}
```

Response Parameters

Field ID	Field Name	Data Type	Description	Example Value
status		Char (10)		success or fail
When status = success				
result		Data	Refer to the Result table below	
When status = fail				
errors		Text		When status = fail

Result

Field ID	Field Name	Data Type	Description	Example Value
merchantId		Char (16)	As per request	
currency		Char (3)	As per request	
amount		Decimal (20, 2)	As per request	
payMethod		Char (45)	As per request	
banks		Array	Refer to the Banks table below	

Banks

Field ID	Field Name	Data Type	Description	Example Value
countryName		Char (45)	Country Name	Thailand
countryCode		Char (2)	Country Code	TH
bankCode		Char (20)	Bank Code	scb
bankName		Char (45)	Bank Name	Siam Commercial Bank

Example

Response Data when the request is success


```
{
  "status": "success",
  "result": {
    "merchantId": "82164414b30aa575",
    "currency": "THB",
    "amount": "200.00",
    "payMethod": "offline",
    "banks": [
      {
        "countryName": "Thailand",
        "countryCode": "TH",
        "bankCode": "kasikornbank",
        "bankName": "Kasikornbank"
      },
      {
        "countryName": "Thailand",
        "countryCode": "TH",
        "bankCode": "scb",
        "bankName": "Siam Commercial Bank"
      }
    ]
  }
}
```


Response Data when request is failed


```
{
  "status": "fail",
  "errors": "There is no bank account available."
}
```

Appendix A - Message Encryption

Encryption Method : AES128
Block Cipher Mode : CBC

Merchant will be provided the followings when merchant account is ready

1) Merchant Key (16 characters)
2) Merchant Hash IV (16 characters)

1) Encrypt the message string (JSON Format) with AES128, cipher mode CBC
2) Use Base64 to encode the AES128 encrypted text

Example

PHP 7

```
$content = '{
      "merchantId": "94b2522ba609909d",
      "transactionId": "20201030023201612",
      "transactionType": "D",
      "payMethod": "offline",
      "currency": "RMB",
      "amount": "500.00",
      "callback": "https://enmu1r0iut2ws.x.pipedream.net",
      "response": "https://enmu1r0iut2ws.x.pipedream.net",
}';

$secret_key = $merchantKey
$secret_iv = $merchantHashIV;
$mode = "AES-128-CBC";
$encrypted = openssl_encrypt($content, $mode, $secret_key, OPENSSL_RAW_DATA, $secret_iv);
$message = base64_encode($encrypted);
```

JAVA 8

```
import javax.crypto.Cipher;
import javax.crypto.SecretKey;
import javax.crypto.spec.IvParameterSpec;
import javax.crypto.spec.SecretKeySpec;
import javax.crypto.IllegalBlockSizeException;
import javax.crypto.BadPaddingException;
import javax.crypto.NoSuchPaddingException;
import java.security.NoSuchAlgorithmException;
import java.security.InvalidKeyException;
import java.security.InvalidAlgorithmParameterException;
import java.io.UnsupportedEncodingException;
import java.util.Base64;

String content = "{" +
  "\"merchantId\": \"94b2522ba609909d\"," +
  "\"transactionId\": \"20201030023201612\"," +
  "\"transactionType\": \"D\"," +
  "\"payMethod\": \"offline\"," +
  "\"currency\": \"RMB\"," +
  "\"amount\": \"500.00\"," +
  "\"callback\": \"https://enmu1r0iut2ws.x.pipedream.net\"," +
  "\"response\": \"https://enmu1r0iut2ws.x.pipedream.net\"" +
"}";

String secretKey = "abcdefghhgfedcba";
String secretIv = "1234567887654321"; //Merchant Pass1 + Pass2

try {
  Cipher cipher = Cipher.getInstance("AES/CBC/PKCS5Padding");
  SecretKeySpec keySpec = new SecretKeySpec(secretKey.getBytes("utf-8"), "AES");
  IvParameterSpec ivSpec = new IvParameterSpec(secretIv.getBytes("utf-8"));
  cipher.init(Cipher.ENCRYPT_MODE, keySpec, ivSpec);
  byte[] ciphertext = cipher.doFinal(content.getBytes("UTF-8"));
  String message = Base64.getUrlEncoder().encodeToString(ciphertext);

  System.out.println("message=" + message);
} catch (NoSuchAlgorithmException e) {
  System.err.println(e.getMessage());
} catch (NoSuchPaddingException e) {
```

```
  System.err.println(e.getMessage());
} catch (InvalidAlgorithmParameterException e) {
  System.err.println(e.getMessage());
} catch (BadPaddingException e) {
  System.err.println(e.getMessage());
} catch (UnsupportedEncodingException e) {
  System.err.println(e.getMessage());
} catch (InvalidKeyException e) {
  System.err.println(e.getMessage());
} catch (IllegalBlockSizeException e) {
  System.err.println(e.getMessage());
}
```

Appendix B - Message Decryption

Decryption Method : AES128
Block Cipher Mode : CBC

Merchant will be provided the followings when merchant account is ready

1) Merchant Key (16 characters)
2) Merchant Hash IV (16 characters)

1) Use Base64 to decode the message to return raw message in bytes
2) Decrypt the raw message in bytes with AES128, cipher mode CBC to obtain the JSON plain data.

Example

PHP 7

```
$request = '{
      "merchantId":"82164414b30aa575",
      "message":"Eis\/kISbtVtkLckK\/LHMQQ1kZ9yy0oZV3tk0CPDJv5xfT+xPvbDT510cPL4b8QedGBsWkHqRmpDfbIGyWZVSB3egVdEEXW6I8q6v
xhccIoHruCclCzrIvP9e\/Y1ick1gkMAEZRCTs5jLLvD3waDU+nzMGIyY2oGtn0V0Lu+bSVAjigRDBrwxnullPU2ABvzh9jq4ARAwxjJQPBVIAAeaWTNsqR
i1dvMo4xp6Oq1i9rUafg4DREkizh\/h4p+1a+YwvluuiMnZC4RcCHsiW5CyjPd+Tg6bSkEYFm4Ddh8bNgLgxxlM9U7BTwX2mvSRnLoZotTgflVAva20LiGc
PeM3AllcPuma5khdlAmxaTnFc3uaoKtWPAPEElgldcEZvh\/2Mlra+2t2MfegpcDpquFwr0+Fm+9ZUg2zQUjAx7ry3N6DuFMlKMTJqxs3\/d\/EXvqI\/gL
8K9ntXMnC229140u71UvZyv+jfDr7mgsSwurW5rsIDpZadwKezEsv7Y7KBe9xGPuh+jNev+SqJKytvoG6xwttnqy+K6IPiLFLwXs54GlhKizDVnIamp2cJp
KeGeH+V7C84jzX5f+dXf+yTDkAffUjpD4b0heJM\/7PMwdZqbFho7KHKZyR9OyHkXcHa2J2jN8FB\/egBh0PWncQutxswGUk1+maQWdZkZM5KzF0SRc50kC
vGZPi2vgvKZMcybnXfOttO2Z+PTmd5b0nOtuo5w=="
}';

$content = json_encode($request);
$content = $content->message;

$secret_key = $merchantKey
$secret_iv = $merchantHashIV;
$mode = "AES-128-CBC";
$encrypted = base64_decode($content);
$message = openssl_decrypt($encrypted, $mode, $secret_key, OPENSSL_RAW_DATA, $secret_iv);
```

JAVA 8

```
import javax.crypto.Cipher;
import javax.crypto.SecretKey;
import javax.crypto.spec.IvParameterSpec;
import javax.crypto.spec.SecretKeySpec;
import javax.crypto.IllegalBlockSizeException;
import javax.crypto.BadPaddingException;
import javax.crypto.NoSuchPaddingException;
import java.security.NoSuchAlgorithmException;
import java.security.InvalidKeyException;
import java.security.InvalidAlgorithmParameterException;
import java.io.UnsupportedEncodingException;
import java.util.Base64;

String requestMessage =
  "dO_ih-LJC7fpBjKRTr0Hbmw0rPvn4MKP3XwDPWxPYFY7T09dEze-tar7NXdYqJ0kY-osQs822lyks-
SI3s9NCidEtbulRDtDeBF2Tq0lhMAGseEDgJnyHu6CtSTLCM3iecKYX9qfsK64GndX1wrIAL2mMPeiWNfYN14rowLliqQU8TkvVR2Rd6Ml2XpecL5YXAUsG
Tq8omxFb3YBB5aYoU9n1rh-NzL_43IQJQw-Tx4Kc-eLniTwNJgVUbpYox3vFjrbVeF3pNFZ832nWnXB7RZ0FLCdz77CUXJ4sQdS17z7JgwXmCPxD22-
YybQixlAbP-tSPIebs1i44qiSo7b9asG7d-y20A7ePk3qgLj_PQ=";

String content = "{" +
  "\"merchantId\": \"94b2522ba609909d\"," +
  "\"message\": \"" + requestMessage + "\"" +
"}";

String secretKey = "abcdefghhgfedcba";
String secretIv = "1234567887654321"; //Merchant Pass1 + Pass2

try {
  Cipher cipher = Cipher.getInstance("AES/CBC/PKCS5Padding");
  SecretKeySpec keySpec = new SecretKeySpec(secretKey.getBytes("utf-8"), "AES");
  IvParameterSpec ivSpec = new IvParameterSpec(secretIv.getBytes("utf-8"));
  cipher.init(Cipher.DECRYPT_MODE, keySpec, ivSpec);
  byte[] requestMessageBytes = Base64.getUrlDecoder().decode(requestMessage);
  byte[] plainText = cipher.doFinal(requestMessageBytes);
  String message = new String(plainText );
```

```
  System.out.println("message=" + message);
} catch (NoSuchAlgorithmException e) {
  System.err.println(e.getMessage());
} catch (NoSuchPaddingException e) {
  System.err.println(e.getMessage());
} catch (InvalidAlgorithmParameterException e) {
  System.err.println(e.getMessage());
} catch (BadPaddingException e) {
  System.err.println(e.getMessage());
} catch (UnsupportedEncodingException e) {
  System.err.println(e.getMessage());
} catch (InvalidKeyException e) {
  System.err.println(e.getMessage());
} catch (IllegalBlockSizeException e) {
  System.err.println(e.getMessage());
}
```